AN EVOLUTIONARY VIEW OF GAME THEORY

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ABSTRACT. Game theory, as seen in class, comprises individual players who make choices and their payoff depends on the choices of all players in the system. Players have a belief about the strategy vectors of other players and reason how they may act. Evolutionary game theory, on the other hand, is defined differently to make it more consistent with evolution. Even if no one is explicitly deliberating or making explicit decisions, game-theoretic concepts continue to apply. These decisions may or may not be made consciously.

Keywords: Game theory, evolutionary game theory, evolution, mixed strategies, evolutionary stable strategies, evolutionary stable mixed strategies, Nash equilibrium, Hawk-Dove game.

1 Evolutionary Game Theory

1.1 Introduction

Evolutionary game theory is the study of the behaviour of large populations of agents who repeatedly engage in strategic interactions. Changes in behaviour in these populations are driven either by natural selection via differences in birth and death rates, or by the application of myopic decision rules by individual agents. Sandholm (2012)

1.2 Evolutionary Biology

The fitness of an organism in an environment is majorly determined by its genes. Fitter organisms produce greater offspring. As a result, genes that produce greater fitness have a higher population representation. This is essentially Darwin's theory of evolution by natural selection.

Evolution is clearly not a one-shot game. Many of the behaviours we see in nature involve numerous organisms in a population interacting. The outcome of these interactions determines how well an organism is going to succeed. We can't measure an individual organism's fitness; it must be assessed in the context of the entire population in which it exists.

This helps us create an analogy with game theory at this point. The *strategy* of an organism is determined by genetically determined traits and behaviour, fitness makes

the *payoff* and *game matrix* is determined by the strategies of the other organisms it interacts with.

1.3 A motivating example

Let us consider a simple example of how the concepts of game theory can be used in evolutionary contexts. Think of a species of beetles. The fitness of a beetle depends on its ability to find and process food effectively. At some point, when the beetles were reproducing, the genes got changed and there was a mutation. We now have a new type of beetles – with larger bodies. These larger beetles consume more food but less of it would go into making them more fit because they would need more food to maintain their metabolism as well. It doesn't look like a very good characteristic to be a large beetle. It seems that these large ones would cease to exist due to natural selection as their fitness decreases. But there is more to this story $\cdots$

Beetles must compete for food and the larger ones at better at claiming an above-average share of food. A payoff matrix denoting a food-related interaction between two organisms can be drawn, the payoffs denoting fitness.

		beetle 2	
		small	large
beetle 1	small	5, 5	1, 8
	large	8, 1	3, 3

From the matrix, we can observe that a faceoff between two beetle of the same type leads to an equal division of food, but the larger beetle gets more food when it faces a small beetle. Large beetles also spend some food in maintaining their body metabolism.

A funny thing to observe here is that none of the beetles are asking themselves "Do I want to be small or large?" We are not dealing with the choice of strategy here, this is happening through generations. Strategy, in this context, is operating over long periods of time due to evolution.

2 Evolutionary Stable Strategies

2.1 Introduction

Nash equilibrium occurs when neither player wants to deviate from their strategy; they are no better off doing something else and it is their best response. But the beetle from Section 1.3 cannot change its strategy! The concept of Nash equilibrium does not work in this setting, rather, a notion analogous to Nash equilibrium, of *evolutionary stable strategies* exist: a genetically wired strategy that, once adopted by a population, continues to stick around.

A strategy is evolutionarily stable if everyone uses it, and any small group of invaders with a different strategy will die off over multiple generations. Easley and Kleinberg (2010)

2.2 Some formalization

Let us consider two strategies S and T. We can say that strategy T invades strategy S at level x (for small x) if a fraction x of the population uses strategy T and fraction $1-x$ of the population uses strategy S. As x is small, $1-x$ is large enough and a larger population uses strategy S than T.

Definition 1. *Strategy S is defined to be evolutionarily stable if there is some number y such that when any other strategy T invades S at any level $x < y$, the fitness of an organism playing S is strictly greater than the fitness of an organism playing T. Hammerstein and Selten (1994)*

Revisiting the beetles from Section 1.3, we would like to check if either being a small or large beetle is an evolutionary stable strategy.

Let for some small number x, a $1-x$ fraction of beetle population is small and x fraction is large. As x is a small number, the population of large beetles is still smaller than that of smaller beetles.

Let us compute the expected payoff being a small beetle in a random food-related interaction. With a probability of $1-x$, a small beetle meets another small beetle resulting in a payoff of 5. With a probability of x, a small beetle meets a large beetle resulting in a payoff of 1. So, the total expected payoff is $5 \times (1-x) + 1 \times x = 5 - 4x$.

Let us compute the expected payoff being a large beetle in a random food-related interaction. With a probability of $1-x$, a large beetle meets a small beetle resulting in a payoff of 8. With a probability of x, a large beetle meets a small beetle resulting in a payoff of 3. So, the total expected payoff is $8 \times (1-x) + 3 \times x = 8 - 5x$.

The expected payoff of Large beetles is $8-5x$ and the expected payoff of small beetles is $5-4x$. For any x, the payoff of being a large beetle exceeds that of being a small one. As a result, being a small beetle cannot be shown to be an evolutionary stable strategy.

If we carry out a similar exercise, assuming that for a small number x, a $1-x$ fraction of the beetle population is large and x fraction is small, then the expected payoff being large is $3 \times (1-x) + 8 \times x = 3 + 5x$. On the other hand, the expected payoff being small is $1 \times (1-x) + 5 \times x = 1 + 4x$. We can observe that for all values of x, the payoff of being large exceeds that of being small, so being a large beetle is an evolutionarily stable strategy.

This game between small and large beetles bears a strikingly close resemblance to the *Prisoner's Dilemma* game we have seen. While the prisoners can choose their strategy (to cooperate or defect), a beetle cannot immediately bring about a transformation to its body size, though evolution over the span of multiple generations can certainly lead to this change.

2.3 A generalized game

We would like to find out some conditions under which a given strategy is evolutionarily stable in a game with generalized payoffs. We will stick to symmetric games, where the roles of the players are interchangeable, as observable in the matrix below.

		organism 2	
		S	T
organism 1	S	a, a	b, c
	T	c, b	d, d

Again, let us assume that for a small number x, a $1 - x$ fraction of the population uses strategy S and x fraction of the population uses strategy T. The expected payoff for using strategy S can be calculated as the sum of interactions with another organism playing strategy S giving the payoff $(1 - x) \times a$ and interactions with another organism playing strategy T giving the payoff $x \times b$. So, the total expected payoff is $a \times (1 - x) + b \times x$. Similarly, the expected payoff for using strategy T is $c \times (1 - x) + d \times x$.

For strategy S to be evolutionarily stable, using Definition 1 the following inequality must hold:

$$a \times (1 - x) + b \times x > c \times (1 - x) + d \times x$$

When x tends to 0, we get the condition $a > c$. For all other values of x, when $a = c$, we get the condition $b > d$.

To summarize, S is an evolutionarily stable strategy in the game we defined when either of the conditions holds: $a > c$ OR $a = c$, and $b > d$.

2.3.1 An intuitive view

We will try to present an intuitive view of the conditions. As S is evolutionarily stable, an organism using strategy S against another organism using the same strategy must get a payoff that is at least as good as using strategy T against strategy S. The condition $a > c$ implies this.

But if strategies S and T both are as good responses to an organism playing strategy S $a = c$, then S is evolutionarily stable only when organisms playing S do better against organisms playing strategy T than two organisms playing strategy T against each other. The condition $b > d$ implies this.

2.3.2 Relations with Nash equilibrium

Let's look at Nash equilibrium in this symmetric game. The strategy pair (S, S) is a Nash equilibrium when S is at least as good response as T, giving rise to the condition $a \geq c$.

The strategy pair (S, S) is a *strict* Nash equilibrium when S is necessarily a better response than T, giving rise to the condition $a > c$.

We can easily state the following two definitions using the previously stated results.

Definition 2. *If a strategy S is evolutionary stable, then (S, S) is a Nash equilibrium.*

Definition 3. *If (S, S) is a strict Nash equilibrium, then the strategy S is evolutionarily stable. Crawford (1992)*

3 Mixed Strategies

3.1 Introduction

There are games in which no strategies are evolutionary stable. The Hawk-Dove game is an example. To understand what unfolds in the Hawk-Dove Game in an evolutionary context, we must broaden the concept of evolutionary stability by permitting some "mixing" of strategies.

3.2 The Hawk-Dove Game

Two animals are fighting over a prize such as a piece of food Leyton-Brown (2008). Each animal can choose between two behaviours, either an aggressive hawk-ish behaviour H, or a gentle dove-ish behaviour D. If one animal is aggressive and the other is gentle, the aggressive animal gains by taking much of the food; but, if both animals are aggressive, the food will be damaged, and they will hurt each other. This leads to the payoff matrix below.

		animal 2	
		D	H
animal 1	D	$3, 3$	$1, 5$
	H	$5, 1$	$0, 0$

We can observe that though there are two pure Nash equilibrium possible, neither hawk nor dove is an evolutionary stable strategy.

3.3 Evolutionary Stable Mixed Strategies

Considering the generalized matrix we saw in Section 2.3, let us calculate the expected payoffs for both organisms 1 and 2.

Organism 1 plays S with probability p and plays T with probability $1-p$. Organism 2 plays S with probability q and plays T with probability $1-q$ Expected payoffs for both organisms 1 and 2, as the situation is symmetric. Probability $p \times q$ of a (S, S) pairing gives a, probability $p \times (1-q)$ of a (S, T) pairing gives b, probability $(1-p) \times q$ of a (T, S) pairing gives c and probability $p \times (1-q)$ of a (T, T) pairing gives d. The total expected payoff can be given as $P(p, q) = p \times q \times a + p \times (1-q) \times b + (1-p) \times q \times c + (1-p) \times (1-q) \times d$.

Definition 4. *p is an evolutionary stable mixed strategy if there is a small positive number y such that when any other mixed strategy q invades p at any level $x < y$, then the fitness of an organism playing p is strictly greater than the fitness of an organism playing q. Easley and Kleinberg (2010)*

Since any evolutionarily stable mixed strategy must correspond to a mixed strategy Nash equilibrium of the game, we will find out a mixed strategy Nash equilibrium in this game. A mixed strategy Nash equilibrium involves at least one player playing a randomized strategy and no player being able to increase his or her expected payoff by playing an alternate strategy. Kang and Rakha (2020)

We assume that (p, p) is a mixed strategy Nash equilibrium. If animal 2 is using a strategy $(p, 1-p)$, the expected payoff for animal 1 from playing Hawk is $5 \times p$, while the expected payoff from playing Dove is $3 \times p + (1-p) \times 1$. Indifference will be created between the two strategies when $5p = 3p + (1-p)$, giving us $p = 1/3$, and the strategy $(1/3, 1/3)$.

From Definition 4, we can say that if p is an evolutionarily stable mixed strategy if q is an invading strategy, for some y and any $x < y$, the following holds

$$(1-x)\ P(p,p) + x\ P(p,q) > (1-x)\ P(q,p) + x\ P(q,q) \tag{1}$$

Here, $P(p, p) = P(q, p)$ for all q played against p. Further, we can compute,

$$P(p,q) = 1/3 \cdot q \cdot 3 + 1/3(1-q) \cdot 1 + 2/3 \cdot q \cdot 5 = 4q + 1/3$$

and

$$P(q,q) = q^2 \cdot 3 + q(1-q) \cdot 1 + (1-q) \cdot q \cdot 5 = 6q - 3q^2.$$

Now, we have

$$P(p,q) - P(q,q) = 3q^2 - 2q + 1/3 = \frac{1}{3}\left(9q^2 - 6q + 1\right) = \frac{1}{3}(3q-1)^2.$$

As this is a perfect square, which is always positive, we can conclude that (1) holds. In other words, it is evolutionarily stable to have a population of $2/3^{rd}$ Hawks and $1/3^{rd}$ Doves.

3.4 Conclusion

The mixed strategy which leads to an evolutionary stable state in the Hawk-Dove game tells us that organisms need to move away from sticking to a particular strategy as it would lead to instability. They need to adopt multiple strategies for their survival.

We can interpret the result of this example in two possible ways: each animal plays the same mixed strategy, or a fraction of the population playing each of the underlying pure strategies in proportion to its contribution to the mixed strategy, i.e., $1/3^{rd}$ of the animals playing Dove and $2/3^{rd}$ playing Hawk.

4 Simulations carried out

Two simulations were carried out to model, analyse and support the statements made in sections 1.3 and 3.2.

4.1 Beetles: small and large

In Section 1.3, we established that being a large beetle is an evolutionary stable strategy. This can be supported using the two plots obtained below.

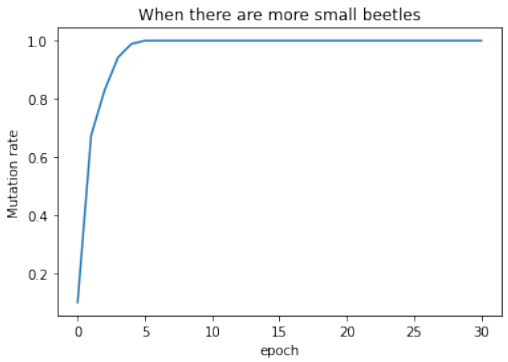


Figure 1: Being small is not an evolutionary stable strategy because a mutation of large beetles would be able to invade and dominate the population.

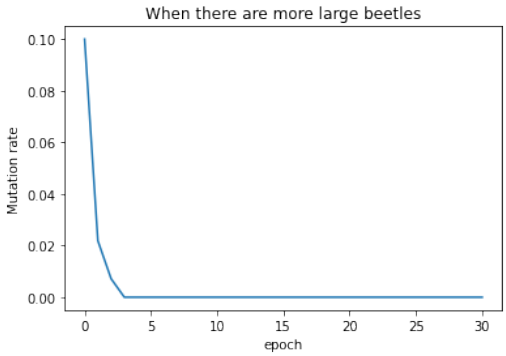


Figure 2: Being large is an evolutionary stable strategy and the mutation by small beetles dies off.

4.2 The Hawk-Dove Game

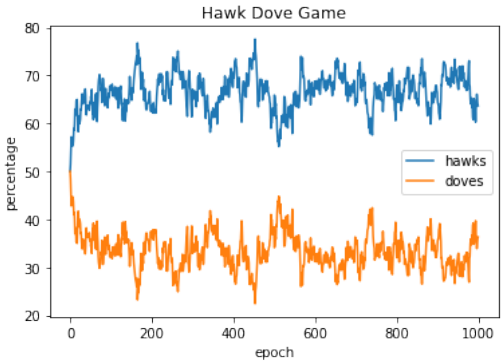


Figure 3: The average percentage of Hawks is 66.56% and that of Doves is 33.44%. This is in line with the (2/3, 1/3) evolutionary stable mixed strategy we derived in the text.

5 Likert-scale ratings

Course Outcome	Relevance
CO1	***
CO2	*****
CO3	-
CO4	*****
CO5	-

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